Approximating Irrational Roots on a Number Line

Bracketing a root between whole numbers, squeezing it to a decimal, and using the estimate in measurement

Directions

- None of these roots is exact, so every answer is an *estimate* — the symbol $\approx$ belongs in your work.
- **Bracket first:** name the perfect square just below the radicand and the one just above it. Their roots are the whole numbers your value sits between.
- **Then squeeze:** square candidate decimals until two of them trap the radicand, and round as the problem asks.
- Measurement answers must carry their unit.

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1. $\sqrt{30}$ lies between the whole numbers ________ and ________.
Name the two perfect squares you used.

2. Give $\sqrt{8}$ to the nearest tenth.

Answer: ________________

3. A square rug covers 45 square feet of floor. How long is one side of the rug, to the nearest tenth of a foot?

Answer: ________________ ft

4. $\sqrt{115}$ lies between the whole numbers ________ and ________.
Name the two perfect squares you used.

5. Give $\sqrt{72}$ to the nearest hundredth.

Answer: ________________

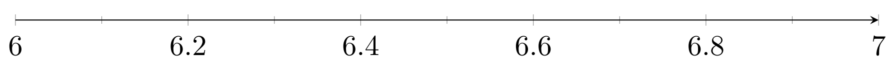
6. A square vegetable bed is being built to cover 150 square feet. How long is one side of the bed, to the nearest tenth of a foot?

Answer: ________________ ft

7. Locate $\sqrt{40}$ on the number line.

(a) To the nearest tenth, $\sqrt{40} \approx$ ________.

(b) Mark that value with a dot on the number line below and label it.



8. Give $\sqrt{95}$ to the nearest hundredth.

Answer: ________________

9. A square tarp must cover a square patch of ground of area 80 square feet. Tarps are sold only in whole numbers of feet on a side. What is the smallest tarp size that covers the patch?

Answer: ________________ ft

10. **Challenge.** Ravi estimates $\sqrt{50} \approx 25$, because half of the number under the radical is 25.

(a) Give $\sqrt{50}$ to the nearest hundredth. Answer: ________

(b) Explain what Ravi did wrong, and show a squaring check that settles it.